

Instructions in preparing an application

Applicants are required to use this template to submit their Flagship program. Minimum font 11pts (Arial, Calibri, Times New Roman), standard line and character spacing, and normal margins. All indicated chapters are mandatory unless otherwise indicated. You must respect the indications given in each chapter in order to adhere to the expected content. A word count is clearly indicated for each section. **If you do not adhere to the maximum amount of words, your application will not be processed.** The word count includes tables, footnotes and figure legends but excludes references. Please do not include hyperlinks. The application form needs to be submitted in a single pdf document, including the budget table and letters of intent. We recommend that you do not send in your submission at the last minute in order to avoid technical difficulties. No submission will be accepted after the deadline, March 15th at 13:00 CET. Applications should be sent in PDF format to Flagships@erasmusmc.nl.

Title: THERANOSTICS

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

<i>Name*</i>	<i>Institution</i>	<i>Department</i>	<i>Email</i>	<i>Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)</i>	<i>For Erasmus MC applicants: clinician/ researcher/ mix</i>	<i>Role in Flagship [Main Lead/Lead/ Applicant]</i>
Yann Seimbille	Erasmus MC	Radiology and Nuclear Medicine	yseimbille@erasmusmc.nl	Tenured position	Researcher	Main Lead
Antonia Denkova	TU Delft	Radiation Science and Technology	a.g.denkova@tudelft.nl	Tenured position		Lead
Lucas Goossens	Erasmus U	Health Policy and Management	goossens@eshpm.eur.nl	Tenured position		Lead
Laura Mezzanotte	Erasmus MC	Radiology and Nuclear Medicine	l.mezzanotte@erasmusmc.nl	Tenured position	Researcher	Applicant
Kristina Djanashvili	TU Delft	Biotechnology	K.Djanashvili@tudelft.nl	Tenured position		Applicant
Rienk Eelkema	TU Delft	Chemical engineering	R.Eelkema@erasmusmc.nl	Tenured position		Applicant
Simone Dalm	Erasmus MC	Radiology and Nuclear Medicine	s.dalm@erasmusmc.nl	Tenured position	Researcher	Applicant
Julie Nonnekens	Erasmus MC	Molecular Genetics	j.nonnekens@erasmusmc.nl	Tenured position	Researcher	Applicant

Mark Konijnenberg	Erasmus MC	Radiology and Nuclear Medicine	m.konijnenberg@erasmusmc.nl	Tenured position	Researcher	Applicant
Gerard van Rhoo	Erasmus MC	Radiation Oncology	g.c.vanrhoon@erasmusmc.nl	Tenured position	Mix	Applicant
Miranda Christianen	Erasmus MC	Radiation Oncology	m.christianen@erasmusmc.nl	Tenured position	Clinician	Applicant
Elizabeth Carroll	TU Delft	Imaging Physics	E.Carroll@tudelft.nl	Tenured position		Applicant
Alina Rwei	TU Delft	Chemical Engineering	A.Rwei@tudelft.nl	Not yet tenured		Applicant
Frederik Verburg	Erasmus MC	Radiology and Nuclear Medicine	f.verburg@erasmusmc.nl	Tenured position	Mix	Applicant
Zoltan Perko	TU Delft	Radiation Science and Technology	z.perko@tudelft.nl	Tenured position		Applicant
Marlies Goorden	TU Delft	Radiation Science and Technology	M.Goorden@tudelft.nl	Tenured position		Applicant
Freek Beekman	TU Delft	Radiation Science and Technology	f.j.beekman@tudelft.nl	Tenured position		Applicant
Esther de Bekker-Grob	Erasmus U	Health Policy and Management	debekker-grob@eshpm.eur.nl	Tenured position		Applicant
Ken Redekop	Erasmus U	Health Policy and Management	redekop@eshpm.eur.nl	Tenured position		Applicant

Corresponding Lead: Dr. Yann Seimbille

Planned start date of seed funding: 1-10-2022

Primary Convergence theme: *Human Centered Technology and AI for Health*

Secondary Convergence theme (if applicable): *Fundamentals of Health & Disease*

Tertiary Convergence theme (if applicable): *Improving Health Journeys*

Quaternary Convergence theme (if applicable): Choose an item.

Keywords: Cancer, Personalized medicine, Radiopharmaceuticals, Nanotechnology, Molecular imaging, AI, Health Technology Assessment

1.b Summary (max 300 words)

Please briefly describe the objectives, design, and anticipated scientific, societal and economic impact for an academic audience.

Theranostic refers to a combination of diagnostics (e.g. imaging) and therapy in one system, which can lead to much better management of patients with oncological diseases. Such a theranostic approach allows for following the treatment and adjusting it accordingly, so that the chance of success is tremendously increased. Theranostics can lead to precision oncology (*treat what you see and see what you treat*) when combined with cancer treatments, such as brachytherapy and hyperthermia. Theranostics provides a fundamental step towards improving survival of patients having 'hard-to-treat cancer types' such as lung, pancreatic and brain tumors, that would highly benefit from the see-and-treat approach.

Despite its high scientific, economic and societal potential, broad clinical applications of theranostics have not been implemented yet. The main reason is the lack of an integrated approach between researchers, clinicians and HTA experts.

The aim of our flagship initiative is therefore: i) to develop bench-to-bedside theranostic technology for diagnosis and treatment of cancer, and ii) to establish a national research program that encompasses the entire technological spectrum of theranostics (from production to clinical implementation).

To achieve these goals, we will develop small molecules, as well as nanoparticles, allowing a theranostic approach, and test them *in vitro* and *in vivo* to bring them a step further to the clinic. In addition, we will implement artificial intelligence (AI) approaches to predict the behavior of these systems. Furthermore, we will develop flexible health economic models, which will guide development of the most promising theranostics from the perspective of value for patients and costs to society.

The Theranostics Flagship allows scientists and clinicians with various expertise from chemistry to economy to join forces, share knowledge and infrastructure in a long-lasting collaboration, leading to new joint projects and much larger network, as well as the establishment of a theranostics institute.

1.c Public Summary (max 300 words)

Layman's summary.

Cancer is indisputably a major burden to western society, causing 3.5 million new cases each year and 1.9 million deaths. To improve the chance of survival for cancer patients, both early diagnostics and appropriate therapies are needed. The term 'theranostics' refers to a combination of therapy and imaging (diagnostics) which allows for tracking the treatment and adjusting it accordingly, so that the chance of success is tremendously increased. Theranostics can lead to precision oncology (*treat what you see and see what you treat*) and help to improve survival of patients having 'hard-to-treat cancer types' such as lung, pancreatic and brain tumors.

However, theranostics are not yet broadly implemented in the clinic. The main reason is the lack of an integrated approach between researchers, clinicians and experts of health technology assessment.

The aim of this flagship project is to develop new and improve current theranostic compounds and evaluate their performance by combining the expertise of researchers from TU Delft, Erasmus MC and Erasmus University. The potential of the theranostics developed by researchers will be evaluated by clinicians and experts of health economics to establish the clinical, economic and societal acceptability of such treatments and their accessibility to patients.

Moreover, this project aims to establish a national research program that will include the entire technological spectrum of theranostics from production to actual clinical implementation, ensuring a long-term impact on cancer treatment. Finally, the theranostic approach is not limited to cancer and during this project we will involve researchers and clinicians working in other medical fields that can benefit from theranostics (for instance cardiology) achieving in this way great scientific and societal impact.

2. Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e

- The Flagship program contributes to the Convergence Health & Technology mission (1)
- The Flagship program objectives fit with the guidance per theme provided by the theme leads, and/or fit with more than one theme of Convergence Health & Technology (2)
- The Flagship program merges approaches and insights from distinct disciplines to reach its goals (3)

- The staff granted by this call actively brings together different disciplines within the program, and creates visibility and attracts talents, partners, and funding for the program (4)
- The Flagship program has a clear view on long term sustainability after the 5-year central funding period (5)
- The envisioned results of the Flagship program have the potential for a significant contribution to healthcare, society and/or have scientific impact which has the potential for long-term societal impact (6)
- The Flagship program provides opportunities for BSc/BA and/or MSc/MA students of the participating institutes to participate in (interdisciplinary) projects (7)
- The proposal defines the pathway of how this contribution can be realized. The pathway includes concrete research and/or impact pathway (societal and/or economical) valorization strategies (8)

2.a Contribution to H&T mission and fit with H&T theme(s)

Theranostics is a combination of the terms **therapeutics** and **diagnostics**. It refers to the coupling of an imaging agent for diagnostics (e.g. radioactive isotope for imaging) to a therapeutic agent within one single system (Fig. 1).¹ A theranostic approach allows for noninvasive monitoring of the treatment and adjusting it accordingly. This way the chance of treatment success amongst patients is tremendously increased compared to conventional treatments. DOTATATE, a radiolabeled peptide developed at Erasmus MC and now marketed as Lutathera® by Novartis for the diagnosis and treatment of neuroendocrine tumors, is the perfect example of how theranostics can have a significant impact on cancer patients' life.² Despite its high scientific, economic and societal potential, applications of theranostics have not yet been broadly implemented in clinical practice. The main reason for that is the lack of an integrated approach between researchers, clinicians and HTA experts.

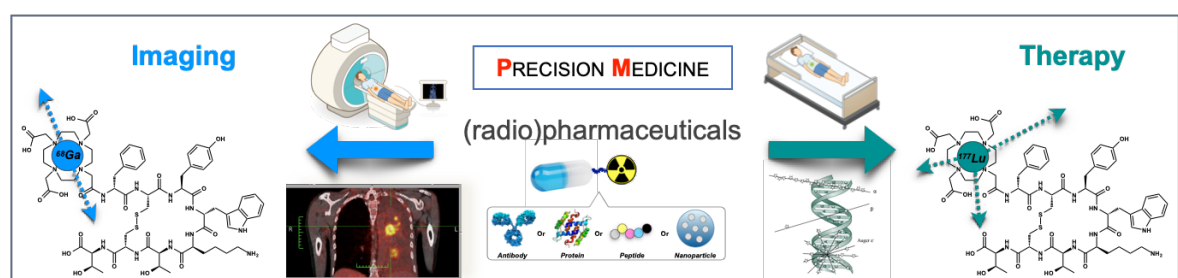


Figure 1. Schematic representation of the theranostic concept.

The aim of this flagship is: i) to develop bench-to-bedside theranostic technology for the diagnosis and treatment of cancer, and ii) to establish a national research program that encompasses the entire technological spectrum of theranostics. I) Development of theranostics will follow a path that starts at the design of molecules, testing them *in vitro* and *in vivo*, predicting their behavior using imaging modalities and AI, and assessing the economic and societal viability through health economic models. II) Simultaneously, the Theranostics Flagship will attract and train the next-generation cross-disciplinary theranostics experts, build an industrial constituency, and thereby create the visibility and driving force to establish a theranostics institute that sustains the efforts that will be made in the coming 5 years. The Theranostics Flagship contributes to three of the four missions and three of the four themes of the Convergence Health & Technology Program.

The goals and objectives of Theranostics directly contribute to **Mission 2**, which envisions **socioeconomic equality in health(care)**. Disadvantaged groups in society show a higher cancer incidence.³ It is well established that the availability of more effective treatments, such as theranostic approaches, has the potential to reduce these socio-economic inequalities in health. Moreover, the affordability of healthcare systems is under pressure given the rising expenditures on health in most countries. This will lead to reduced coverage of care, reduced accessibility of health insurance, and financial problems particularly for people in disadvantaged positions. Health economic research can help steer the development of theranostics towards treatments with the

most potential to deliver value for money. For these reasons, our flagship includes expertise on health economics with the aim to carefully consider **the economic and societal acceptability** of theranostic treatment and their **accessibility** to patients.

Theranostics Flagship also contributes to **Mission 3**, related to **prevention and early diagnostics to proactively maintain health**. Theranostic approaches allow performing **early cancer diagnostics and selecting treatments that are much more patient friendly** using the same molecules for both purposes.⁴ One example is using nanoparticles with a dual functionality that are on one hand MRI contrast agents for diagnosis and imaging of disease progression, and on the other hand can be applied for brachytherapy (internal radiotherapy) combined with hyperthermia.^{5,6} Such a combined treatment needs just a single visit to the hospital reducing the burden to the patient. Moreover, it can be primarily applied in the initial stages of the disease, hence the importance of early diagnostics. This example as well as others in our proposal help to **proactively maintain health**. Health will be maintained for a longer period in the population if patients can be diagnosed and treated at the early-stage. Therefore, merging diagnostics with therapeutics will likely improve future cancer care.

Theranostics connects to **Mission 4**, concerning **Individually tailored precise medical treatment for all**. Theranostic lends itself extremely well to **precision oncology**: targeted therapies within a theranostic framework allow oncologists to treat what they see and see what they treat. A theranostic approach helps to improve patient selection due to information obtained by imaging, allowing the creation of **individual-based treatment plans**. For example, selection of patients for the most appropriate radionuclide therapy regimen (e.g. lutetium-177-PSMA-617) is achieved by the imaging of adequate radiotracer uptake in all tumor sites in order to deliver a radiation dose sufficient to accomplish therapeutic goals.⁷

2.b Synergy between disciplines

Elaborate on how this program merges approaches and insights from distinct disciplines to reach its goals.

We expect that this flagship will bring forth a new treatment technology by developing theranostic agents, thereby providing an array of treatments for cancer that will become available to patients. The significance of theranostics is its versatility to be used in all oncological applications and not being limited to one type of cancer. This requires well-coordinated contributions from the disciplines that are represented in this proposal: biology, chemistry, physics, mathematics, medicine and health economics.

Biology lies at the basis of the selection of potential treatments that have the promise to be beneficial to patients and help to understand why certain treatments do not work, so that better therapeutic approaches can be developed. Chemistry is needed to synthesize all the different theranostic molecules, while physics and mathematics are essential to analyze the obtained imaging data and to predict the behavior of the theranostics (e.g. through AI). To eventually implement the theranostic agents in the clinic the disciplines of medicine and economy are needed. Throughout the process, health economic models will enable selection of the most promising theranostic treatments for further development, by balancing health gains and costs and calculating the cost-efficiency of treatments based on a minimally needed health gain and/or maximally acceptable health costs.

The flagship partners have the expertise needed to make this proposal a success ranging from synthesis of small molecules and nanoparticles to analysis of imaging data and development of various imaging modalities and therapies.

2.c Long term sustainability plan

Please indicate your view on the feasibility of the program as well as the long-term sustainability plan of the program after the 5-year central funding period. Describe possible risks and how to tackle them.

Feasibility of the program is ensured by:

- **Quality of the research team:** a multidisciplinary researcher team with established track records in the theranostic field (see chapter 4).
- **Continuous learning cycle:** the experiences and know-how gained from developing and introducing theranostic treatments with a high TRL level (e.g. theranostic molecules) can be leveraged when initiating new theranostic concepts and initiatives at a low TRL (photon activated systems).
- **Excellent research infrastructure:** unique research facilities (e.g. research nuclear reactor, small animal imaging facility) in the vicinity of the largest university medical center in the Netherlands (Erasmus MC).
- **Advisory board:** an international Advisory Board will monitor the quality of the program and advise on scientific progress, societal impact and management issues.

Long-term sustainability after the 5-year central funding period is safeguarded by:

- A **business/sustainability plan** will be created, detailing exploitation strategies for the created results, including information about potential protection of the results, spin-off companies and follow-up projects.
- **Training program:** PhD students, post-docs and mid-career researchers will be provided with the best possible research and professional skills to empower them to become the next generations of highly competent researchers, scientific leaders, and entrepreneurs in the field of theranostics.
- **Continuity of leadership and organization:** During the 5 years of the Theranostics program, main lead Yann Seimbille and co-leads will act as its scientific leaders. In order to support career development of other consortium members, they will be succeeded by mid-career scientists from year six onwards. Theranostics senior scientists will carefully track the development of the mid-career scientists during the first five years in order to ensure a smooth transition and sound continuation of scientific leadership.
- **Consortium expansion:** Main focus during the first 5 years is on including academic and industrial partners in the field of oncology. After that, the consortium will gradually be expanded by actively involving other researchers, HTA experts, clinicians and industry partners focused on theranostic solutions to build towards a full-fledged theranostics institute.

Risks and mitigation strategies

Risk management is a continuous process and a key aspect of the overall management of the program. A dedicated risk management system (RMS) is set up and maintained by the Management Board and the scientific staff of the program for monitoring of the actual risks and measures taken. All research projects on which the Flagship embarks will face yearly Go/No-Go evaluations by the Advisory Board using the following criteria: (1) goals reached in the 1st period, (2) outlook and chance evaluation for success of new theranostic treatments, and (3) impact of the activity for the consortium and society. For projects that face major setbacks or prove not viable, the scientific staff

(together with the Advisory Board) will decide on the project's contingency plan to define routes to either tackle emerging scientific and bottlenecks or to embark on highly challenging but promising new directions.

At this stage a number of potential risks is classified into categories, taking into account impact and probability, as well as their contingencies (see table 1). Most of the risks that could cause a problem are either at a low or medium level and have limited impact.

Table 1: Project risks and the proposed mitigation measures. L=low, M=medium, H=high (in both likelihood (LH) and impact).

Source	Description of risk	Risk	Impact	Mitigation strategies
Researchers	Insufficient number of high-quality candidates for PhD student or postdoc positions	L	M	Involvement and exposure of Theranostic in MSc programs. Collaborations with national and international research schools to attract qualified juniors/PhD students, job advertising at various international (social) media platforms.
	PhD student or postdoc leaves during the project	L	M	Quick procedures and reserve funds at Erasmus Uni, EMC, and TU Delft to hire new staff; Clear data management procedures for easy transfer of research.
Leading Staff	Withdrawal of leading staff from program	L	M	Succession via replacement schemes by similarly qualified staff, including current tenure-track positions in the consortium.
	Conflict	L	M	Clear division of responsibilities; Conflict resolution procedures are in place, involving clear escalation procedures, ultimately to the formal responsible persons, being the deans for personnel affairs, finances, etc. and the Management Board for the scientific aspects.
Operational	Financial drawbacks, unforeseen costs	M	H	Appointment of financial administrator within the administrative team; Close monitoring of financial status.
Scientific	Absences of key persons for long period of time	M	M	Each WP will be coordinated by a PI in close collaboration with the scientific management team. If necessary, other PIs involved in the program can take over tasks.
	Breakdown of key equipment	L	M	For many types of key equipment multiple versions are available at the partner's labs. It is a strength of Theranostics that researchers can use equipment at other locations while structural solutions are found for down-time of key equipment.
	Program overtaken by developments elsewhere	L	L	Risk is low, as we are at the forefront in all relevant research fields and can quickly incorporate the newest scientific insights into our research.
	WP is not successful	M	H	Redirect to different activity. Theranostics is set up in such a way that the WPs are not strictly dependent on each other. The HTA element of the project will help avoiding development of treatments that are not economically feasible. In case proposed theranostic compounds still fail, the acquired knowledge would still be useful to feed into the development of the AI models.

2.d Visibility, attracting talents and new partners

Describe how you will create visibility for the flagship program, how you will engage students (BSc/BA and/ or MSc/MA) and early career researchers in interdisciplinary projects, as well as how you plan to attract talents, partners and funding.

Visibility for the flagship program

To create visibility for the Theranostics Flagship, a communication, dissemination and exploitation (CDE) plan with stakeholder-specific strategies will be created. Different channels will be used, including open access publication in scientific journals, stakeholder specific meetings, symposia, workshops & trainings, social media, press releases, project website and newsletters.

Engagement of students

BSc/BA and MSc/MA students are regularly conducting their theses at the groups of the consortium members. This would allow to incorporate their theses into the flagship project. BSc and MSc courses and guest lectures on Theranostics will be organized to increase students' awareness on this specific field of research and attract them for their internships.

Plan to attract talents

The Theranostics Flagship aims at attracting the most talented early career researchers in the theranostics field. We will actively scout students via (selective) Master's program at the three partner institutions, our international networks, and by worldwide advertisements. For example, advertisements will be placed on EURAXESS, prominent journals (e.g. Nature Jobs) and social media (LinkedIn).

Plan to attract partners

Dedicated Theranostics Industrial Platform Meetings will be organized to attract industrial partners to join the Flagship, and drive utilization and valorization activities emanating from our program. Company representatives are invited to join these meetings, where scientific staff will present scientific outcomes of the Flagship program. In turn, company representatives are requested to provide scientific and career advice to PhD students and postdocs. The Industrial Platform aims to inspire entrepreneurial scientists with the establishment of their start-up companies. We already have and will further establish direct communication lines with industrial partners and draw upon the expertise of the valorization offices of the three institutes in this. The attendance of physicians is very important for the transition from lab to practice since they can share their positive experience in the clinic and feedback with their peers, but also to show a tangible outcome of the important research being performed. Furthermore, we will apply for funding programs specifically aimed at stimulating public-private partnerships.

Plan to attract funding

The Theranostics Flagship will attract co-funding through applying for national and international funding programs over the course of the 5-year. The consortium will organize at least two Funding Meetings per year together with the grant offices of the three institutes to identify funding programs suitable for co-funding this Flagship. Co-funding for the first two years is already ensured (see budget table). Other funding programs that we intend to apply for later in the Flagship program are for instance ERC, NWO and KWF.

2.e Long term societal impact

Describe the potential and possible pathways for a significant contribution to healthcare, society and/or scientific impact which has the potential for long-term societal impact. Pathways include concrete research and/or impact pathway (societal and/or economical) valorization strategies. If applicable, indicate expected TRLs and necessary partners. In case of fundamental research please describe the scientific impact and time scale of expected societal impact.

The societal impact is two-fold: (i) the project will lead to more effective and more personalized treatments of patients and (ii) it will reduce the need for animal experiments. Small molecules and nanoparticles will be developed with the dual purpose of individualized diagnosis and treatment and then be preclinically tested. Combining this information with imaging data and AI models will help to predict the behavior of theranostics yet to be discovered, and therefore a better assessment can be made before such molecules enter into the phase of *in vivo* testing (i.e. reducing animal experiments). To achieve a scientific impact, the results of these projects will be shared with peers through journal publications, conferences and workshops.

Different theranostic compounds will be the main product of this project which are expected to have an important societal impact since they will allow for early diagnosis, as well as follow-up and adjustment of the course of treatment to improve chance of success. As the TRL levels of the theranostics in this project vary, different pathways will be required. For theranostics that are already in an advanced stage the next step will be to enter into clinical trials, while the other compounds will require pre-clinical studies. To achieve societal and economic impact it is important to valorize these theranostic compounds through patents, as well as start-up companies, so that the next step in the product's development can be achieved.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

- The Flagship program contributes to excellent Convergence Health & Technology research, at the forefront of international research (1)
- The Flagship program is both innovative as well as challenging in its research and approach (2)
- The Flagship program has a clear strategy to engage students in its research plan (3)
- The Flagship program has a clear aim and long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Coherence within the program is clearly demonstrated (4)
- The Flagship program aligns with or builds on (inter)national research agendas in this field, fit with the (inter)national research agendas but has a “high risk, high gain” main objective that is thoroughly explained (5)
- The Flagship program is feasible in terms of research performed, staff and budget needed (6)
- The Flagship program has a concrete plan for attracting external funding, which should result in at least the same amount as the central funding over the 5-year period (7)

3.a Ambitions

Describe how the ambitions of the flagship program are in line with Convergence Health & Tech research. Please elaborate on the innovative and challenging aspects of the research program and/or its approach.

The Theranostics Flagship contributes to three of the four themes of the Convergence Health & Technology Program. All the WPs are aimed at fueling the convergence knowledge pipeline on theranostics, connecting and translating basic research to new theranostic modalities against cancer.

These modalities provide early cancer detection, diagnosis, treatment monitoring and precision medicine, all within one system. It aligns exactly with the aim of **Theme 1: Fundamentals of Health & Disease** to produce new knowledge and technology, and contribute to its translation into actionable diagnostic information, reliable therapeutic decisions, and ethically acceptable, societally supported solutions to healthcare challenges.

The Theranostics Flagship will lead to precise delivery of therapeutic agents to maximize efficacy while minimizing side effects. This approach involves the imaging of the drugs at the tumor site, followed by the therapy. This personalized approach aligns very well with **Theme 2: Improving Health Journeys**, which expects development of novel imaging technology for a pro-active and personalized whole-life approach to health and healthcare.

Lastly, our current healthcare system needs to transform from a ‘disease cure system’ into a connected system that focuses on the entire health journey throughout life, as outlined in **Theme 3: Human-centered technology & AI for health**. This theme aims to promote individual health and implement effective prevention strategies, early disease detection and more personalized treatment from the earliest moments and throughout life. Theranostics connects to this aim by delivering novel diagnostic-therapy combinations against oncological diseases that can occur at different stages of life.

The main innovative aspect of the proposed research program is the inclusion of various technology (e.g. small molecules, nanoparticles, AI, PDT, radionuclide therapy), while the main challenge is to orchestrate the different activities to compose the perfect theranostic “symphony”. Thanks to the complementary expertise involved, all different aspects of the theranostic development, from bench-to-clinic, are taken into consideration.

3.b Aims and feasibility

Describe the program’s aims, including a clear strategy to involve students in its research plan. Elaborate on the program’s feasibility in terms of research performed, staff and budget needed. Address a concrete plan for attracting external funding. Also address possible risks and how to tackle these.

Aim

The field of theranostics is still at its infancy and it will undoubtedly grow towards other applications and diseases beyond nuclear medicine, such as image-guided surgery or photodynamic therapy. Building on previous successes at EMC (e.g. DOTATATE, NeoBOMB1) and joining efforts with TU Delft and EUR experts, we aim i) to **develop novel bench-to-bedside theranostic technology** for oncological diseases, and ii) to **establish a center of excellence** that encompasses the entire technological spectrum of theranostics.

Strategy to involve students

Bachelor and Master students will be invited to perform their research internships in the groups of the consortium members to support the flagship projects. Participation to different national events (e.g. Workshop of the Dutch Radiochemical Society, Chains) will help to attract in talented students.

Feasibility of the program is ensured by:

- Strong scientific foundation of breakthrough discoveries in the field of theranostics in the last two decades:
 - Development at EMC of DOTATATE and NeoBOMB1 (now in Novartis’ portfolio).^{2,8,9}
 - Development of nanoparticles for combined brachytherapy and hyperthermia supported by Elekta.¹⁰
- Quality of the scientific staff (see 4.a-b).
- Long-term experience in acquiring funding from competitive (inter)national grant programs (see 4.a).

Staff and budget needed:

One PhD student per WP will be required for WP1, WP2 and WP3. A bench fee of 20k euro per year per student is required for the synthesis, labeling and preclinical testing of the theranostics. Two postdocs for 2 years are planned for WP3, to work on the light sheet imaging system and AI. A PhD student and a postdoc (6 months for student’s supervision) will be dedicated to the WP4. We estimate that a budget of approximately 400k euro per year for 5 years will be needed for the Theranostics Flagship.

Plan to attract external funding:

- Institutional business development/valorization offices will support the WP leads acquiring external co-funding from companies (e.g. PerkinElmer, Novartis, Bayer, Telix)
- Two Funding Meetings per year will be organized with the institutional grant offices to identify funding programs suitable for co-funding and prepare a bi-yearly funding roadmap (see 2.d).
- Theranostics Industrial Platform Meetings will be organized to attract industrial partners to join the Flagship, drive utilization and valorization activities emanating from our program, and secure external funding (see 2.d).

Possible risks and mitigation:

A dedicated risk management system (RMS) will be set up and maintain to manage risks and according mitigation strategies (extensively described in 2.c).

3.c Work plan

Elaborate on the long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Pay attention to overall coherence of the work plan.

WP1 - Design of Nanosystems for Theranostics (M1-M48)

Leader(s): Kristina Djanashvili (TUD) and Laura Mezzanotte (EMC)

Participants: Antonia Denkova (TUD), Rienk Eelkema (TUD), Alina Rwei (TUD), Yann Seimbille (EMC), Gerard van Rhooen (EMC) and Miranda Christianen (EMC)

Objectives: Design of two theranostic nano-systems: **(O1.1)** nanoparticles for MRI-guided hyperthermia and open-source brachytherapy, and **(O1.2)** image-guided triggered drug delivery using polymeric nano-carriers.

Approach: (Task 1.1) The MRI-guided hyperthermia/brachytherapy research will build on a joint project between Erasmus MC and TU Delft in which the nanoparticles have been produced and characterized.¹⁰⁻¹² The project will further exploit the full potential of the methodology to result in a clinically viable product. **(Task 1.2)** Drugs administered using conventional delivery systems often exhibit low efficacy and side effects. To address this challenge, target-specific and controlled drug delivery needs to be investigated. Therefore, we aim to develop ROS (reactive oxygen species)-responsive polymeric nanomaterials to trigger the release of targeted drug and/or photodynamic (PDT) agents.¹³⁻¹⁷ ROS will be generated by either ionizing radiation or light activation of PDT agents. Image-guided drug delivery will be achieved by the incorporation in the nano-formulations of near infrared emitting PDT agents.

Deliverables: (D1.1) Nano-particles for combined brachytherapy and hyperthermia, **(D1.2)** New polymeric systems for image-guided drug release triggered by ROS.

Contribution to project: This WP will help to develop combined therapies to improve efficiency of current cancer treatments. This WP is related to WP3 for the biological evaluation of these systems.

WP2 - Small molecules for theranostics (M1-M48)

Leader(s): Yann Seimbille (EMC) and Antonia Denkova (TUD)

Participants: Simone Dalm (EMC), Julie Nonnekens (EMC), Mark Konijnenberg (EMC), Frederik Verburg (EMC), Kristina Djanashvili (TUD) and Rienk Eelkema (TUD)

Objectives: Based on results of existing TUD/EMC collaboration (Convergence Kick-off), we will develop libraries of low molecular weight theranostics to target cancer cells and tumor microenvironment (TME) and select for lead candidates **(O2.1)**.¹⁸⁻²⁰ We will turn these small-molecules sequentially into imaging **(O2.2)** and therapeutic agents **(O2.3)** for theranostic purposes.

Approach: (Task 2.1) Peptide Libraries will be synthesized and screened via high-throughput assays for lead identification. **(Task 2.2)** Studies to determine binding affinity, stability, uptake and retention in cells and human cancer tissues will allow selection of the most promising molecules for *in vivo* studies. **(Task 2.3)** Our small-molecule based theranostics will be radiolabeled with gamma emitter for SPECT and PET imaging, and with β^- emitter or alpha emitting radionuclide for targeted radionuclide therapy (TRT). Labeling of the lead compounds will be performed through a bioorthogonal approach to ensure selective attachment of the radiolabeled tag to the peptides.²¹ **(Task 2.4)** This chemical approach may also be explored for pretargeting or the preparation of dual-labeled compounds for preoperative diagnosis and image-guided surgical resection of tumor lesions.^{22,23} All needed infrastructures are readily available at the three participating institutes.

Deliverables: (D2.1) A new generation of peptide-based theranostic targeting the TME and the malignant cells.

Contribution to project: This WP will identify small molecules suitable for the dual purpose of theranostics. There will be close collaboration with WP3 to study and validate the small molecules in an *in vitro* and *in vivo* setting.

WP3 Biological evaluation & Prediction performance (M1-M60)

Leader(s): Simone Dalm (EMC) and Zoltan Perko (TUD)

Participants: Elizabeth Carroll (TUD), Marlies Goorden (TUD), Freek Beekman (TUD), Laura Mezzanotte (EMC), Mark Konijnenberg (EMC), Frederik Verburg (EMC) and Julie Nonnekens (EMC)

Objectives: The most promising theranostics (WP1&2) will be evaluated using state-of-the-art preclinical *in vitro* and *in vivo* setting **(O3.1)**, imaging modalities **(O3.2)** and AI methods **(O3.3)**.

Approach: (Task 3.1) Model creation, *in vivo* image optimization (timing and mass), *ex vivo* confirmation using autoradiography and/or fluorescent imaging will be carried out. Preclinical therapy and preliminary toxicity studies will be performed to determine efficacy and safety of the novel therapeutics.²⁴ Biological evaluation of the novel (radio)therapeutic will also include light sheet imaging **(Task 3.2)** and computational approaches **(Task 3.4)**.^{25,26} Thanks to the generation of a considerable amount of imaging and biodistribution data, AI methods will be developed to predict therapeutic outcome. AI neural networks will be trained on retrospective animal or cell data. Such AI models will be used to optimize experimental design and imaging protocols, and to reduce the number of animals needed in pre-clinical testing.

Deliverables: (D3.1) Biodistribution and therapy efficacy studies, **(D3.2)** development of light sheet imaging technique, and **(D3.3)** AI-based methods to predict therapeutic outcome.

Contribution to project: this WPs will evaluate the translational potential of the theranostics developed in WP1&2 and guide the selection of theranostics for IMPD submission and subsequent clinical trials.

WP4 Economics of the proposed theranostic approaches (M13-M60)

Leader: Lucas Goossens (EUR)

Participants: Esther de Bekker-Grob (EUR), Ken Redekop (EUR) and Frederik Verburg (EMC)

Objectives: (O4.1) we will study the economic and societal viability of theranostics using Health Technology Assessment (HTA).^{27,28}

Approach: The economic and societal acceptability of theranostics depends on the balance of the benefits and costs. The discipline of Health Technology Assessment (HTA) studies the consequences of medical interventions. It assesses the value of health improvements for patients and society, and shows trade-offs that are involved. Relevant to this proposal, a method called 'early HTA' can examine the viability of theranostics, even before it can be clinically evaluated, by exploring the limits and the desirability of the potential costs and effects. This requires close cooperation with WP1&2 and can give guidance for choices made in these WPs. **(Task 4.1)** We will develop flexible and adaptable health economic models to calculate the maximally acceptable costs given assumed health benefits, as well as the minimally required benefits given certain cost assumptions. **(Task**

4.2) In addition to the early HTA, we will apply expertise from the Erasmus Choice Modelling Centre to investigate preferences among patients and in society at large for different aspects of theranostics and similar future health innovations.

Deliverables: (D4.1) Flexible health economic models for the assessment of theranostic treatments. (D4.2) Studies of cost-effectiveness and patient preferences during and after the development of treatments.

Contribution to project: This work package will help ensure that theranostics developed in the previous WPs are societally feasible and desirable.

Gantt Chart

Smart Chart																								
	2022				2023				2024				2025				2026				2027			
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
WP1. Design of Nanosystems for Theranostics																								
M1.1																								
M1.2																								
WP2. Small molecules for theranostics																								
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WP3. Biological evaluation & Prediction performance																								
M3.1																								
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M3.3																								
WP4. Economics of the proposed approaches																								
M4.1																								
M4.2																								

3.d Alignment of the program with relevant national and international research agendas

Explain how the program aligns with or builds on national and international research agendas in its fields or, if it does not fit with these agendas, elaborate on what is the “high risk, high gain” objective.

With the recent growth of novel cancer therapies, the theranostic field with its strategy - *if you can see it, you can treat it* - represents an important link towards precision health solutions. Therefore, this flagship program is perfectly in line with the Dutch Science Agenda (NWA) and in particular the route ‘Gezondheidszorgonderzoek, preventie en behandeling’. This route strives to answer questions that fall directly in the scope of the Flagship aims such as: i) how do we make healthcare as high-quality as possible but keep it affordable? and ii) how can we identify new targets for molecular therapies?

It also addresses the goals defined in the implementation plan of the European mission on cancer to improve the lives of over 3M people by 2030. This plan is based on better diagnosis and treatment of cancer. Moreover, the content of this flagship tackles the objectives of Europe’s Beating Cancer Plan that sets out new approaches to the entire disease pathway and recognizes the importance of comprehensive cancer infrastructures. Theranostic centers are rapidly being set worldwide (Molecular Imaging and Theranostics Center in Missouri/USA, Stanford Center of Theranostic Care, Theranostics Center of Molecular Radiotherapy in Bad Berka/Germany). Notably, there is no similar center in The Netherlands, which does not mean that theranostics are underrepresented in our country, but rather emphasizes the need for timely support for better structuring and facilitation of this research field in order to position it at the cutting edge.

To boost the development of theranostics, this flagship proposes to develop and combine key technologies on bio-related and soft materials, nanomedicine and imaging technologies, as well as

application of AI to reduce the number of lab animals needed in pre-clinical studies in alignment with the 3Rs (replacement, reduction and refinement) defined by the EU regulation policy on animal exploitation.

4. Consortium (25%) max 2000 words for 4.a to 4.d

- The Flagship program is driven and executed by a well-organized team of scientists from all three institutions and at least 4 departments or faculties in total. In exceptional cases, it is possible to form a Flagship program with only two leads from two institutions, but this should be clearly motivated (1)
- Members of the consortium have a track record of excellence in academic research, translation of scientific results into clinical application or societal impact (2)
- Members of the consortium have a significant track record in engaging and supervising BSc and MSc students in research projects (3)
- Members of the consortium have demonstrated to be able to attract external funding, thereby enhancing the programs sustainability after the 5 years of central funding (4)
- There is a demonstrable fit between the aims and objectives of the program and the expertise of the consortium members (5)
- End users (private and/or public sectors) are involved from an early start and team members are well-connected to external partners in the ecosystem (6)
- Governance should be clearly described and is organized in such a way that it facilitates collaboration and optimally makes use of the expertise of all participants (7)

4.a Track record of the consortium members

Elaborate on the consortium members' track record in academic research, translation of scientific results into clinical application or societal impact, BSc and MSc supervision and engagement in research projects, ability to attract external funding after the 5-year central funding period.

Erasmus MC, Department of Radiology and Nuclear Medicine

Yann Seimbille (Associate Professor, leader RadioPharmaceutical Chemistry) - Main lead	
Expertise	Development of theranostics and multimodality imaging probes
Grants, awards	Approx. €1.6M (KWF, Kansen voor West and Convergence) in the last 5 years
Supervision	Currently, 2 Postdocs, 5 PhD students, 4 MSc students
Career highlights	Clinical translation of [²²⁵ Ac]Ac-PSMA-I&T

TU Delft, Department of Radiation Science and Technology

Antonia Denkova (Associate Professor, leader Applied Radiation and Isotopes) - Lead	
Expertise	Soft matter and radiochemistry
Grants, awards	€1.8 M granted (last 5 years); Aspasia award 2020
Supervision	Currently, 1 Postdoc, 6 PhD students and 40+ MSc students. Past: 8 successful PhD students
Career highlights	Coordinator of Marie Curie ITN (Innovative Training Network) project called Trace'nTreat

Erasmus University, Erasmus School of Health Policy and Management

Lucas Goossens (Assistant Professor, Master program director) - Lead	
Expertise	Health technology assessment, health economics, epidemiology

Grants, awards	Approx. €1.8M of grants (Dutch hospital association and ministry of Health, ZonMW and Longfonds); Value in Health Paper of the Year Award; ESHPM Education Award
Supervision	Currently, 3 PhD students. Past: >20 MSc students
Erasmus MC, Department of Radiology and Nuclear Medicine	
Laura Mezzanotte (Associate Professor, leader Genetic Engineering for Multimodality Imaging)	
Expertise	Multimodality imaging of immune cells; theranostic nanoparticles; probes for image-guided surgery and PDT
Grants, awards	€2.0M (EU grants, KWF, NWO, TKI); Acquisition of a MSOT system for preclinical research
Supervision	Currently, 2 Postdocs, 4 PhD students. 3 PhDs have defended their theses
Career highlights	Council member of the European Society of Molecular Imaging
Simone Dalm (Assistant Professor, leader Radiotracer Interactions)	
Expertise	Evaluation of novel radiotracers; mechanism of action of radiotracers; combination treatment strategies
Grants, awards	KWF Young Investigator's Grant, ZonMw Veni, Alavi Mandell award, Erasmus University Research Prize
Supervision	Currently, 2 Postdocs, 4 PhD students, 2 technicians
Career highlights	Pre-clinical validation and optimization of NeoB
Mark Konijnenberg (Research Physicist)	
Expertise	Dosimetry and dose-effect relations for targeted radionuclide therapies; dosimetry models from cellular to small animal and clinical therapy
Career highlights	Member of the EANM oncology and theranostics committee
Frederik Verburg (Full Professor, head of Translational Nuclear Medicine)	
Expertise	Early-phase testing of theranostic concepts; development of novel approaches for radionuclide therapy and treatment of thyroid cancer
Grants, awards	> €1 M in research funding (Daniel den Hoed foundation, KWF) since 2020
Erasmus MC, Department of Molecular Genetics	
Julie Nonnekens (Assistant Professor, leader Radiobiology of radionuclide therapy research)	
Expertise	DNA damage repair mechanisms; radiation biology of targeted radionuclide anticancer treatment; combination treatment strategies
Grants, awards	Approx. €4.8M of grants in the past 5 years; ERC starting grant
Supervision	Currently, 2 Postdocs, 6 PhD students, 2 MSc students and 1 technician
Erasmus MC, Department of Radiation Oncology	
Gerard van Rhoon (Full Professor)	
Expertise	Radiotherapy, hyperthermia
Career highlights	President of the European Society for Hyperthermic Oncology; senior editor of the Int. J. of Hyperthermia

Miranda Christianen	
Expertise	Radiation oncology, brachytherapy
Grants, awards	>35 publications in high-impact journals
TU Delft, Department of Biotechnology	
Kristina Djanashvili (Associate Professor)	
Expertise	Synthesis and evaluation of novel MRI contrast agents; smart combination of multimodal imaging and radiotherapy/hyperthermia
Grants, awards	NWO (Veni, HTSM-NWO-TTW), Horizon-2020 (€1.2M); >55 publications
Supervision	5 Postdocs, 7 PhD students, 40+ MSc students.
Career highlights	Management committee of COST Action "Theranostics: Imaging and Therapy and Bimodal PET-MRI Molecular Imaging Technologies"
TU Delft, Department of Radiation Science and Technology	
Zoltán Perkó (Assistant Professor)	
Expertise	Computational physics, deep learning, uncertainty management tools and therapy plan optimization
Grants, awards	>€1.7M from TU Delft, NWO Veni, KWF, HollandPTC-Varian grant, SAMOSAfer, Euratom Horizon2020, ASSAS Euratom
Freerk Beekman (Full Professor)	
Expertise	Development of imaging detectors, image reconstruction and imaging systems (SPECT, PET, Optical Imaging, X-ray CT)
Grants, awards	16 patents; various awards (Delft Entrepreneurial Scientist 2010; Frost & Sullivan Product Innovation 2011; FOM Valorization 2013; Innovation of the Year 2015 & 2018; Edward Hoffman Memorial Award 2017; Hasegawa Memorial 2021)
Marlies Goorden (Assistant Professor)	
Expertise	SPECT-PET-CT imaging; software for iterative image reconstruction
Career highlights	Optimization of gamma ray detectors by using novel detection algorithms, various scintillating materials and new combinations of sensors
TU Delft, Department of Chemical Engineering	
Rienk Eelkema (Associate Professor)	
Expertise	Design, synthesis and application of responsive small molecules and polymer materials
Grants, awards	NWO (Veni, Vidi), ERC (Consolidator), FOM, H2020; > 90 publications in peer-reviewed journals
Supervision	>15 PhD students and postdocs, and >50 MSc/BSc students
Career highlights	Two patents in diagnostics application
Alina Rwei (Assistant Professor)	
Expertise	Photo-responsive drug delivery systems

Career highlights	>20 publications in high-impact journals
TU Delft, Department of Imaging Physics	
Elizabeth Carroll (Assistant Professor)	
Expertise	Optics and microscopy
Grants, awards	Delft Technology Fellowship; ZonMW; NWO, Delft AI labs.
Erasmus University, Erasmus School of Health Policy & Management	
Esther de Bekker-Grob (Full Professor)	
Expertise	Health preference research, health economics
Supervision	Currently, 4 PhD students
Ken Redekop (Associate Professor)	
Expertise:	Early health technology assessment of diagnostics, devices, precision medicine
Grants, awards	Six recent grants and ongoing EU Horizon projects.
All applicants have extensive experience in supervising MSc/BSc students.	

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Demonstrate the fit between the aims and objectives of the program and the expertise of the consortium members. Also describe the complementarity of the members.

The Theranostics Flagship brings together a multidisciplinary team of researchers that act at the forefront of their respective fields. They rely on the latest advances in drug development, chemistry, biology, nanotechnology, nuclear medicine, precision medicine, physics, radiology and radiation oncology to develop bench-to-bedside theranostic technology for diagnosis and treatment of cancer. The researchers of **Erasmus MC** are led by Yann Seimbille (WP1&2), who is an expert on radiopharmaceutical sciences. Particularly, his knowledge of developing theranostics and multimodality imaging probes and the clinical translation thereof will be instrumental for generating peptide-based radiopharmaceuticals for imaging and treatment of cancer. **TU Delft** under the lead of Antonia Denkova (WP1&2) provides expertise within nanotechnology, radiochemistry and brachytherapy. With this expertise, she will lead the research on the nanosystems-based theranostics. WP3 includes the biological evaluation of the theranostics designed in WP1&2, as well as some technological developments (light sheet imaging, AI) to predict the therapeutic outcome. The lead of this WP will be shared among the researchers having the required expertise: Simone Dalm (EMC) and Zoltan Perko (TUD). The research group of **Erasmus University** under the lead of Lucas Goossens (WP4) will complement the team with knowledge on health technology assessment, health economics and epidemiology to study the desirability of the new theranostic approaches amongst patients, society at large and policy makers.

The consortium also contains a generation of early and mid-career biomedical researchers, engineers, societal scientists and clinicians. Their complementary expertise is essential for the success of the current program and continuation thereafter. It is organized in the work package configuration presented in table 2 to stimulate cross-overs and collaboration.

Table 2: Work package configuration for cross-over of disciplines – WP1: Design of nanosystems for theranostics; WP2: Small molecules for theranostics; WP3: Biological evaluation & prediction performance; WP4: Economics of the proposed theranostic approaches.

Name (Institution)	Expertise	WP1	WP2	WP3	WP4
Laura Mezzanotte (EMC)	Nanoparticle, MRI, optical imaging	X		X	
Kristina Djanashvili (TUD)	Nanoparticle, MRI	X	X		
Rienk Eelkema (TUD)	Organic and polymer chemistry	X	X		
Gerard van Rhoon (EMC)	Hyperthermia	X			
Miranda Christianen (EMC)	Radiation oncology	X			
Alina Rwei (TUD)	Drug delivery systems, optogenetics	X			
Simone Dalm (EMC)	Tracer evaluation		X	X	
Julie Nonnekens (EMC)	Radiobiology, radionuclide therapy		X	X	
Mark Konijnenberg (EMC)	Dosimetry		X	X	
Frederik Verburg (EMC)	Nuclear medicine, clinical translation		X	X	X
Elizabeth Carroll (TUD)	Light imaging			X	
Zoltan Perko (TUD)	AI, modeling			X	
Marlies Goorden (TUD)	Biomedical imaging			X	
Freek Beekman (TUD)	Biomedical imaging			X	
Esther de Bekker-Gron (EUR)	Preference measurement, choice modeling				X
Ken Redekop (EUR)	Health technology assessment, early HTA				X

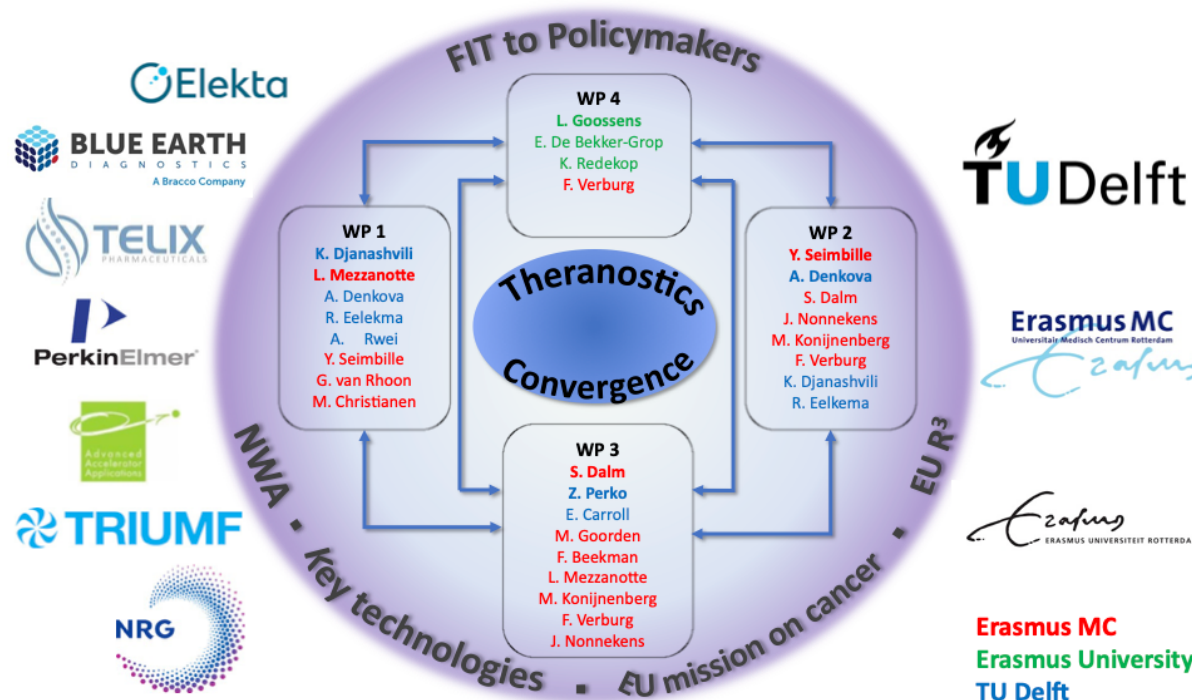


Figure 2. Complementarity and interactions of the full consortium.

4.c Involvement of partners

Indicate which partners are involved, how and in what stage they are involved, as well as how well team members are connected to the external partners in the ecosystem.

Industrial partners contribute their expertise and products for translating, validating and upscaling theranostics to a clinical setting. This knowledge and expertise is still scarce and not limited to Dutch borders. Therefore, both national and international partners are involved in the consortium. **Nuclear Research and consultancy Group (NRG) (NL)** is a leader in the development of nuclear medicines (medical isotopes). NRG deploys its expertise to research and support nuclear medical treatments, particularly for personalized and targeted cancer therapy. **TRIUMF (Canada)** is a particle accelerator centre and a partnership among leading Canadian research universities. TRIUMF provides centralized resources, tools and expertise in pursuit of pioneering science from the search for the smallest particles available to the development of new technologies, including medical isotopes. TRIUMF will support the Flagship consortium by providing isotopes and space for collaborative work. Besides its expertise, TRIUMF also serves as a best practice of a nationwide academic collaboration to facilitate frontier research.

Alfarim (NL) is a company specialized in the industrialized production of actinium-225 based on bombardment of radium-226 with protons generated by a medical cyclotron. Alfarim's particular contribution to this consortium is the supply of 10-20 mCi of Ac-225.

PerkinElmer (United States) is a leading, global provider and manufacturer of measuring equipment for research in areas such as diagnostic healthcare, pharmacy, food safety, biotechnology and industrial applications. PerkinElmer closely collaborates with commercial, government, academic and healthcare customers to deliver and develop reagents, assays, instruments, and strategic services. PerkinElmer will support the consortium with training in optical imaging technology and technical assistance for *in vivo* fluorescent and bioluminescent imaging studies.

4.d Governance, organization and collaboration strategy of the consortium

Please describe how the consortium is organized to facilitate collaboration and make optimal use of the expertise of all participants.

The Theranostics Flagship is an ambitious program that requires a decisive and transparent (flat) organizational structure based on direct communication and goal-oriented management to enhance progress.

Project organization

The **management board** (lead applicants) conducts the daily management, the administrative and financial aspects. The **scientific management team** (lead applicants & WP leaders) will be responsible for interacting with the Flagship members and external partners, overseeing the WPs, determining the overall strategy, research and training activities, risk management, visibility for recruiting talented researchers (e.g. recruitment days, symposia, workshops), outreach activities and (public) dissemination, and attracting co-funding. The scientific team will have monthly meetings to discuss progress and strategy of the program. The **advisory board** (representatives from co-funders, external partners and applicants, and external scientific advisors) will monitor the quality of the project and advise on scientific progress, societal impact and management issues, and identify new opportunities. The Advisory Board will also actively interact with PhD students, postdocs and early-career researchers, in order to share their knowledge and experience, e.g. through active participation in the annual Theranostics symposia and Industrial Platform Meetings. The Advisory Board will meet every year at the symposia and will draw a report on the progress and strategy.

Project governance

To ensure high-quality monitoring of the project, a data management plan (DMP) according to the FAIR principles will be set-up at the start of the project, including mandatory project report dates. There will be regular assessment of whether or not the planned milestones and deliverables of the project have been achieved. If delays occur, the management board will decide together with the scientific management team whether a contingency plan must be implemented.

Organization of scientific life & collaboration strategy

The scientific management team and management board will have monthly meetings to discuss and communicate past performance, plans for the upcoming year as well as contingency plans and developmental needs to assure optimal productivity and coherence of the program. The WP leaders will have regular progress meetings with their scientists, and discuss the collaboration and progress of the WP together with all WP participants in quarterly WP-meetings. To facilitate cross-talk between work packages, addressing challenges or research and translational questions, cross-WP meetings will be scheduled twice a year. The advisory board will be consulted for advice on the progress of the Theranostics Flagship, particularly with respect to 'go/no-go' decisions on research lines and the choice of new directions.

A vibrant scientific life within the consortium is vital and will be achieved by organizing different types of activities. Activities include a kick-off, mid-term and final conference, two-day symposia organized for all the Theranostics Flagship members to present their results; the general scientific program of the conferences and symposia will be open for external participants and serve to boost the scientific exchange, the research progress and discussion of plans.

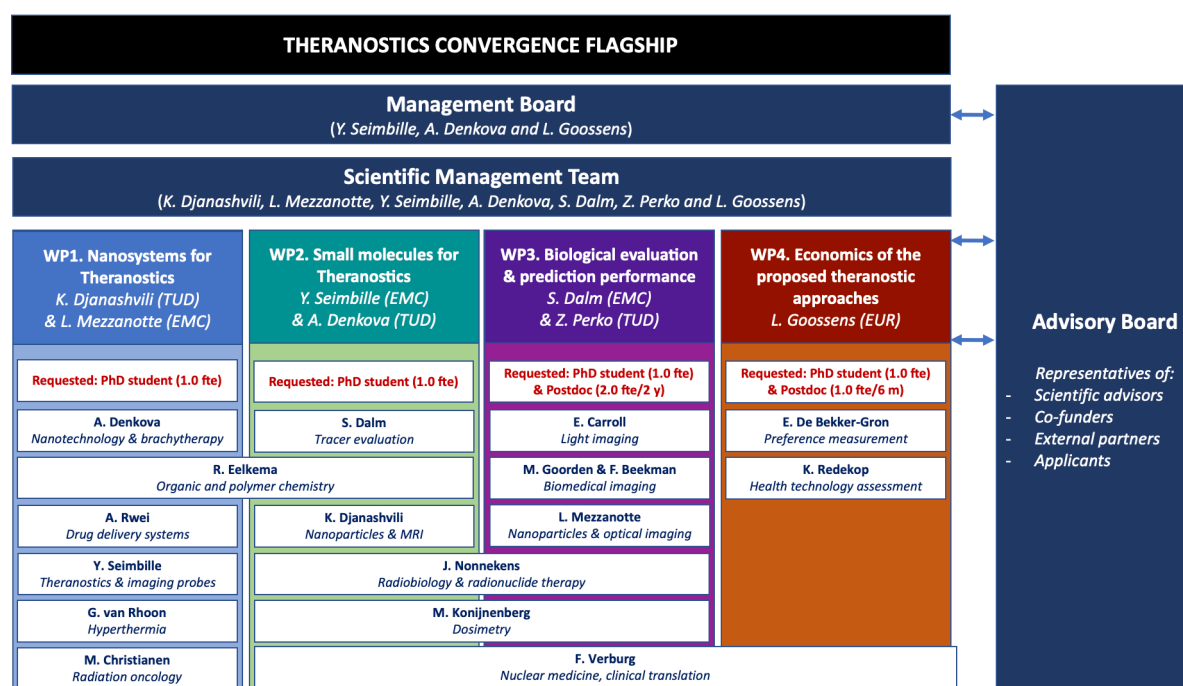


Figure 3. Organizational structure of the consortium.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

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Budget formulier: Convergentie Flagship Call

Flagship:	Thermostics
Flagship acronym:	Enigma MC
Naam Pernoorder:	01-09-2022 t/m 31-8-2027
Startdatum en looptijd van het project:	5 jaar
Totaal project budget:	€3.853.111

5 jaar	60 maanden	Partners
budget Flagship	€4.000.000,00	Totaal € 3.853.111

EMC	€ 2.246.480
TUD	€ 462.435
EUR	€ 310.176
Totaal	€ 3.019.091

R&D Overview: kostensoorten per type onderzoek (hele project periode)

Locatiekosten	Partner die de kosten maakt	Functie	Aantal fe	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten	2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
									in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoeksorganisaties																							
	EMC	PHD	1,0	01.09.22	31.08.26	Schaaft 10.5	€ 3.788	€ 273.456	€ 22.788		€ 88.364	€ 88.364	€ 88.364	€ 45.576		€ 45.576					€ 273.456	€ 0	€ 273.456,00
	EMC	PHD	1,0	01.09.22	31.08.26	Schaaft 10.5	€ 3.788	€ 273.456	€ 22.788		€ 88.364	€ 88.364	€ 88.364	€ 45.576		€ 45.576					€ 273.456	€ 0	€ 273.456,00
	EMC	PHD	1,0	01.09.23	31.08.27	Schaaft 10.5	€ 3.788	€ 273.456	€ 22.788		€ 88.364	€ 88.364	€ 88.364	€ 45.576		€ 45.576					€ 273.456	€ 0	€ 273.456,00
	EMC matching	postdoc	1,0	01.09.22	31.12.23	Schaaft 11	€ 4.000	€ 97.920	€ 24.480		€ 73.440		€ 88.364								€ 97.920	€ 0	€ 97.920,00
	EMC matching	PHD	1,0	01.09.23	31.08.27	Schaaft 10.5	€ 3.788	€ 273.456	€ 22.788		€ 88.364	€ 88.364	€ 88.364	€ 45.576		€ 45.576					€ 273.456	€ 0	€ 273.456,00
	EMC matching	PHD	1,0	01.09.23	31.08.27	Schaaft 10.5	€ 3.788	€ 273.456	€ 22.788		€ 88.364	€ 88.364	€ 88.364	€ 45.576		€ 45.576					€ 273.456	€ 0	€ 273.456,00
	EMC matching	postdoc	1,0	01.01.24	31.08.27	Schaaft 11	€ 4.000	€ 289.260													€ 289.260	€ 0	€ 289.260,00
	TUD	postdoc	1,0	01.09.22	31.08.24	PostDoc TUD	€ 4.209	€ 151.519	€ 25.253	€ 25.253		€ 75.760	€ 50.508	€ 73.440	€ 73.440	€ 50.508	€ 73.440	€ 48.960	€ 0		€ 151.519	€ 0	€ 151.519,22
	TUD	postdoc	1,0	01.09.24	31.08.26	PostDoc TUD	€ 4.209	€ 151.519				€ 75.760	€ 50.508			€ 50.508				€ 151.519	€ 0	€ 151.519,22	
	TUD matching	postdoc	1,0	01.01.23	30.09.23			€ 49.556			€ 49.556									€ 49.556	€ 0	€ 49.556,50	
	TUD matching	Techniciën	1,0	01.01.23	31.12.23			€ 10.631			€ 10.631									€ 10.631	€ 0	€ 10.631,00	
	TUD matching	PHD	1,0	01.09.23	31.08.27	PHD TUD	€ 3.102	€ 223.335			€ 19.611	€ 55.834	€ 55.834	€ 55.834	€ 37.223					€ 223.335	€ 0	€ 223.335,00	
	EUR	PHD	1,0	01.09.22	31.08.26	Schaaft 10.5	€ 3.788	€ 273.456	€ 22.788		€ 88.364	€ 88.364	€ 88.364	€ 45.576							€ 273.456	€ 0	€ 273.456,00
	EUR	postdoc	1,0	01.09.22	28.02.23	Schaaft 11	€ 4.000	€ 39.700	€ 24.400		€ 12.400										€ 39.700	€ 0	€ 39.700,00
Ondernemingen																							
	Onderneming 1						€ -	€ 0													€ 0	€ 0	€ 0,00
	Onderneming 2						€ -	€ 0													€ 0	€ 0	€ 0,00
	Onderneming 3						€ -	€ 0													€ 0	€ 0	€ 0,00
	Onderneming 4						€ -	€ 0													€ 0	€ 0	€ 0,00
Overige (privé) partners																							
	Naam 1						€ -	€ 0													€ 0	€ 0	€ 0,00
	Naam 2						€ -	€ 0													€ 0	€ 0	€ 0,00
	Naam 3						€ -	€ 0													€ 0	€ 0	€ 0,00
	Naam 4						€ -	€ 0													€ 0	€ 0	€ 0,00
Totaal								€ 2.811.216													Total locatiekosten	€ 2.811.216	

Kosten van materialen en hulpmiddelen:			2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
Partner die de kosten maakt	Materiaal / hulpmiddel	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind		
Onderzoeksorganisaties	EMC	Biochips (synthesis, in vitro & in vivo testing and labelling of the thermostics)	€ 240.000		€ 48.000		€ 48.000		€ 48.000		€ 48.000,00		€ 32.000,00		€ 240.000		€ 240.000,00	
	EMC matching	Consumables and materials	€ 80.000	€ 5.333	€ 16.000		€ 16.000		€ 16.000		€ 16.000,00		€ 10.666,67		€ 80.000	€ 0	€ 80.000,00	
	TUD	Consumables	€ 30.000	€ 2.000	€ 6.000		€ 6.000		€ 6.000		€ 6.000,00		€ 4.000,00		€ 30.000,00	€ 0	€ 30.000,00	
	TUD	Laptop	€ 4.000	€ 267	€ 800		€ 800		€ 800		€ 800,00		€ 533,33		€ 4.000,00	€ 0	€ 4.000,00	
	TUD matching	Materials	€ 15.600		€ 7.800		€ 7.800								€ 15.600,00	€ 0	€ 15.600,00	
Ondernemingen	Onderneming 1														€ 0	€ 0	€ 0,00	
	Onderneming 2														€ 0	€ 0	€ 0,00	
	Onderneming 3														€ 0	€ 0	€ 0,00	
	Onderneming 4														€ 0	€ 0	€ 0,00	
Overige (privé) partners	Thermostics	Material testing full time 100K dollar per year	€ 450.000		€ 50.000,00	90000		€ 90.000,00		90000		€ 90.000,00		€ 40.000,00		€ 450.000	€ 0	€ 450.000,00
	Thermostics	EMC grant Acc225 totaal 150.000	€ 150.000		€ 30.000,00	30000				30000		€ 30.000,00		€ 20.000,00		€ 150.000	€ 0	€ 150.000,00
	Naam 1														€ 0	€ 0	€ 0,00	
	Naam 2														€ 0	€ 0	€ 0,00	
Totaal			€ 969.600												Totaal kosten materialen en hulpmiddelen		€ 969.600	

Kosten van gebruik van machines en apparatuur					2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
Partner die de kosten maakt	Machine / apparatuur			Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind		
Onderzoeksorganisaties																				
	Organisatie 1																€ 0	€ 0	€ 0,00	
	Organisatie 2																€ 0	€ 0	€ 0,00	
	Organisatie 3																€ 0	€ 0	€ 0,00	
	Organisatie 4																€ 0	€ 0	€ 0,00	
Ondernemingen																				
	Onderneming 1																€ 0	€ 0	€ 0,00	
	Onderneming 2																€ 0	€ 0	€ 0,00	
	Onderneming 3																€ 0	€ 0	€ 0,00	
	Onderneming 4																€ 0	€ 0	€ 0,00	
Overige (privé) partners																				
	Naam 1																€ 0	€ 0	€ 0,00	
	Naam 2																€ 0	€ 0	€ 0,00	
	Naam 3																€ 0	€ 0	€ 0,00	
	Naam 4																€ 0	€ 0	€ 0,00	
Totaal						€ 0												Totaal kosten machines en apparatuur		€ 0

Aan derden verschuldigde kosten			2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	Naam derde, omschrijving kosten	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	
Onderzoeksinstituten	Organisatie 1														€ 0	€ 0	€ 0,00
	Organisatie 2														€ 0	€ 0	€ 0,00
	Organisatie 3														€ 0	€ 0	€ 0,00
	Organisatie 4														€ 0	€ 0	€ 0,00
Ondernemingen	Onderneming 1														€ 0	€ 0	€ 0,00
	Onderneming 2														€ 0	€ 0	€ 0,00
	Onderneming 3														€ 0	€ 0	€ 0,00
	Onderneming 4														€ 0	€ 0	€ 0,00
Overige (privé) partners	Naam 1														€ 0	€ 0	€ 0,00
	Naam 2														€ 0	€ 0	€ 0,00
	Naam 3														€ 0	€ 0	€ 0,00
	Naam 4														€ 0	€ 0	€ 0,00
Totaal		€ 0													Totaal kosten derden		€ 0

5.	Publicatie, reis- en verblijfskosten		2022		2023		2024		2025		2026		2027
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To
THERANOSTICS
Flagships Proposal Group

Name Ron Koop
Telefon + 49 6423/5429-775
Email Ronald.Koop@perkinelmer.com

Rodgau, 28.02.2022

Subject: Letter of intent for training and technical support

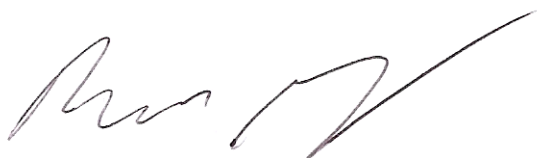
To whom it may concern

This is to declare our intention to support the 'THERANOSTICS Flagships Proposal Group' with technical assistance and training.

PerkinElmer is a major provider of instrumentation and reagents for in vivo optical imaging. We have a group of imaging experts in Europe that can offer technical support for our customers.

We will support the THERANOSTIC consortium by training members and students to use optical imaging technology to track fluorescent theranostic agents in vivo. We will furthermore provide technical assistance for planning, optimizing and performing in vivo fluorescent and bioluminescent imaging studies.

Thank you,



Ron Koop, Ph.D. | Subject Matter Expert, In Vivo Imaging
PerkinElmer LAS (Germany) GmbH

Beratung und Verkauf

Telefon 0800 – 181 00 32
Fax 0800 – 181 00 31

Service

Telefon 0800 – 000 66 79
Fax 0800 – 588 86 81

D-63110 Rodgau-Jügesheim
Ferdinand-Porsche-Ring 17
Telefon (06106) 610-0
Fax (06106) 610-487

March 12th, 2022

Prof. Yann Seimbille
Professor and Head of Radiochemistry,
Department of Radiology and Nuclear Medicine
Erasmus Medical Centre
Rotterdam, The Netherlands
y.seimbille@erasmusmc.nl

Dear Dr. Seimbille,

This letter is to indicate my enthusiastic support for application toward the Convergence Flagship Call titled: "Theranostics". Given your history of productive research, demonstrated leadership in radiochemistry, and the team you have assembled for this opportunity, I do believe that you are uniquely positioned to deliver on the proposed research.

I note your team is composed of experts that span disciplines, ranging from scientists to clinicians, all with experience and interests that range from basic discovery through to the applied. With Convergence funding, this group will be poised to share knowledge and infrastructure that will significantly enhance interactions and knowledge transfer and development between several institutions.

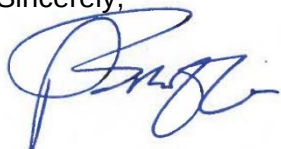
Your application includes a proposal to develop novel radiotheranostic approaches by researching, developing, and translating radiolabeled small molecules as well as larger constructs, such as nanoparticles, and test them in preclinical models to bring them closer to the clinic. Beyond the basic science, you propose to leverage emerging tools such as artificial intelligence (AI) to help predict the behavior of these radiopharmaceutical systems, and further develop flexible health economic models to guide development of the most promising theragnostic drug leads – all with a goal of enhancing precision medicine for patients, ultimately modernizing healthcare. I anticipate that the proposed effort across the team of experts you have brought together will create lasting collaborations between all organizations involved, but also have significant scientific and long-term societal impact.

With that I wish to convey TRIUMF's full support for your efforts as it is highly aligned with our own mandate of nurturing a dynamic isotope and radiopharmaceutical research and development program based-in Vancouver, but with international focus. TRIUMF is enabled by five onsite cyclotrons that produce a wide variety of medical isotopes on a daily basis, several of which are relevant to your efforts. With these machines we can provide your group with routine access to radioisotopes, such as ^{227}Th , ^{225}Ac , ^{213}Bi and ^{212}Pb , along with many more companion diagnostic isotopes, including ^{203}Pb , ^{89}Zr , ^{64}Cu , etc. TRIUMF can also provide visiting researchers access to advanced, high-level radiochemistry facilities, access to skilled personnel and isotopes that are difficult to transport (such as cyclotron-produced ^{68}Ga).

In all, I estimate TRIUMF's in-kind support in the provision of isotopes and space for collaborative work to be worth between \$30,000 and \$100,000/year depending on the level of interaction, and look

forward to working with you to define specific, mutually beneficial efforts between our institutions upon securing your Convergence funding.

Sincerely,

A handwritten signature in blue ink, appearing to read "Paul Schaffer", with a stylized, cursive script.

Paul Schaffer, Ph.D

Director, Life Sciences Division, TRIUMF

Associate Professor, Radiology, University of British Columbia

Adjunct Professor, Chemistry, Simon Fraser University

CHEVATO CAPITAL PARTNERS HOLDING BV

Attn: Dr Hanno Mak
Bloemcampaan 10
2244 EC, Wassenaar
The Netherlands

Address
Molengraaffsingel 12
2629 JD Delft
The Netherlands

Telephone
+31 (0)15 744 0101

Email
info@alfarim.com

Website
www.alfarim.com

Subject
Letter of Intent for TU Delft -
Flagship program round 2021-
2022 – Theragnostics Project

From
Hanno Mak

Our reference
EB/HM/RID/2022-03

Date
14 March 2022

Dear Miss Denkova, dear Antonia,

I, Hanno Mak, in my capacity of Chief Executive Officer (CEO) at AlfaRim Medical BV hereby confirm that AlfaRim intends to contribute to the Theragnostics project, as applied for by the main applicant, Yann Seimbille, Researcher at Erasmus MC.

AlfaRim will produce GMP-grade Ac-225 through the Ra-226(p,2n)Ac-225 reaction using a small medical cyclotron supplying Ac-225 free from Ac-227 contamination. For that purpose, AlfaRim is currently developing a state-of-the-art radiopharmaceutical production facility on the Campus of the TU in Delft. Ac-225 is one of the rarest medical isotopes in the world and current market prices vary from €5.000-€10.000 per millicurie (mCi) of non-GMP-grade Ac-225. It's AlfaRim's intention to support your project by a.o., supplying 10-20 mCi of GMP-grade Ac-225.

AlfaRim intend to contribute at a total of 150K€ in-kind (supply of Ac-225, permitted that our facility is operational at the time of the research project) to this proposal. With the full proposal we will then submit a support letter specifying our contribution to the research proposal.

Yours sincerely,



Hanno Mak



To whom it may concern

Direct dial 010-7035987

Fax number 010-7034033

Internal postal address Ne-507k

E-mail a.vanderlugt@erasmusmc.nl

Our reference 20220311ALYS

Date March 11, 2022

Concerning: Letter of support for the Theranostics Flagship application

Postal address

P.O. box 2040
3000 CA Rotterdam, NL

Dear Sir/Madam,

Visiting address

Dr. Molewaterplein 40
3015 GD Rotterdam, NL

I am writing this letter to strongly support the application of leaded by Dr. Yann Seimbille for the Convergence Flagship project on Theranostics.

Contact & route
www.erasmusmc.nl

Theranostic is a concept in which a molecule can be used sequentially as an imaging agent and a therapeutic. Erasmus MC has been a pioneer in this field with the discovery of DOTATATE, a peptide targeting the somatostatin receptor SSTR2, which was granted FDA and EMA approval for imaging and treatment of neuroendocrine tumors in 2017. Theranostics have completely revolutionized nuclear medicine, and more and more applications of this concept are now reaching other specialities. Many research avenues are therefore yet to be explored. The proposed Flagship will gather a multidisciplinary research team with established track record in the theranostic field to bring this concept a step further. Know-how and expertise in chemistry, biology, physics, health sciences and economics will be combined to push the actual boundaries towards a wider application of theranostics in clinical practice. Furthermore, the location for this program is optimal. Indeed, the researchers will benefit from access to unique research facilities (e.g. research nuclear reactor, small animal imaging facility) in the vicinity of the largest university medical center in the Netherlands, allowing to realize all the steps of the molecule-to-man paradigm. The ambition of the Theranostic Flagship is not only to develop technology for a better diagnosis and treatment of cancer, but also to establish a research program that will train and help career growth of the next generation scientists. Ultimately, the objective will be to set-up a theranostic center where state-of-the-art research, education and healthcare will be provided.

Yours sincerely,

Aad van der Lugt, M.D., PhD
Professor and Chairman of the department of
Radiology and Nuclear Medicine